

ADAM D. RUSHDY

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EDUCATION

Purdue University - B.S. Aeronautical & Astronautical Engineering, Honors

Expected Graduation: May 2028

GPA: 3.82/4.00

Activities: Purdue Aerial Robotics Team (R&D Airframe), Teaching Assistant, Purdue All-American Marching Band

EXPERIENCE

Engineering Intern - *Xchanger*

Hopkins, MN | May 2026 - Present

- Tested pressure-loss performance of air-to-air heat exchangers across 9 flow conditions using the Log-Tchebycheff traversal method (18 points/measurement), improving data-acquisition accuracy by ~10% and identifying a ~3x underestimation in predicted pressure-drop, enabling model recalibration.
- Generated performance curves for 44% of air-air unit configurations by combining empirical airflow and differential pressure data.
- Configured experimental airflow test sections with minimum 8 duct diameters upstream and 3 downstream of sampling locations, minimizing turbulence effects and improving measurement reliability.

R&D Airframe Member - *Purdue Aerial Robotics Team (PART)*

West Lafayette, IN | Sep 2025 - Present

- Designed and manufactured 9 wing structure and tail assembly components for a team eVTOL aircraft focused on modularity and easy assembly with SolidWorks.
- Developed MATLAB structural analysis to evaluate rib spacing trade-offs between panel buckling and structural weight, informing airframe design decisions.
- Developed CAM toolpaths for CNC machining of 2 precision carbon-fiber layup wing molds within a 2 hour manufacturing constraint using Fusion 360.
- Fabricated 2 composite wing assemblies using carbon fiber layup over laser-cut internal structures.

Teaching Assistant - *Honors Intro to Engineering (ENGR 16X)*

West Lafayette, IN | Aug 2025 - Present

- Support 60+ first-year students with coursework and grading including 6 design projects spanning Python modeling, MATLAB simulation, and Raspberry Pi embedded systems.

Intern - *Seminar for Top Engineering Prospects (STEP)*

West Lafayette, IN | Jun - Aug 2025

- Developed and delivered a modular Arduino programming curriculum to 320 prospective engineers, built functions and tutorials to teach students basic programming skills.

Teaching Assistant - *Intro to C Programming*

West Lafayette, IN | Jan - May 2025

- Led 30+ students across 12 labs, held weekly office hours covering debugging and core C concepts, and graded weekly coding assignments.

PROJECTS

9-State Inertial Navigation System - In Progress

June 2026 - Present

- Developing a 9-state INS with Python integrating ESP-32, IMU, GPS, and magnetometer for 6-DOF position and velocity estimation at 40 Hz, benchmarking against flight controller logs to target position error under 5 meters.
- Implemented complementary and Kalman filter sensor fusion across 9 sensor streams, targeting drift correction to under 20 m/min during GPS outages and attitude error under 5° in all axes.

Fixed-Wing Surveillance UAV - Design, Analysis & Fabrication

Mar - June 2026

- Designed, analyzed, and fabricated a 1.3 m, 1.7 kg fixed-wing UAV with ~30 fps raspberry pi video integration. Selected airfoils in XFOIL, sized wing geometry for ~16 m/s cruise, and modeled the full assembly in SolidWorks.
- Performed aerodynamic, stability, and flight dynamics analysis in XFLR5, iterating on CG, tail moment arm, and wing geometry to achieve a 12.5% static margin.
- Designed wing structure for a 3g load factor; fabricated using carbon-fiber skin layups over 3D-printed internal ribs. Built weight of 1.7 kg vs. 1.5 kg predicted, with mass growth attributed to composite skin uncertainty.
- Built a parallel foam-board testbed, validating flight controller implementation before maiden flight and catching 4 configuration issues.

SKILLS

CAD / Analysis: SolidWorks, Fusion 360 (CAM), NX, Teamcenter, XFLR5, XFOIL, Simulink

Programming: MATLAB, Python, C/C++, Java, Arduino, Git/GitHub

Fabrication: Carbon-fiber composite layup, 3D printing (FDM), CNC milling

Additional: Eagle Scout, BSA Troop 479 (2017–2023) — 150+ volunteer hours in construction, conservation, and team leadership.